Summary Notes

Overview

Here is an overview of the topics covered in this session:

1. Session 1
    a. Configuring Azure Resources using the Azure Portal, PowerShell and CLI
    b. Comparison of Administrator Tools
    c. Working with Azure Resource Manager (ARM) Templates
    d. Introduction to Azure Bicep Files

2. Session 2
    a. Understanding Virtual Networks in Azure
    b. Planning and Creating Virtual Networks
    c. Subnet Creation and Configuration
    d. Configuring Public and Private IP Addressing
    e. Network Security Groups (NSGs) and Security Rules
    f. Application Security Groups (ASGs)
    g. Azure DNS and Domain Hosting

Session 1

Configuring Azure Resources Using Azure Portal, PowerShell and CLI

- **Azure Portal**: A web-based interface for managing Azure resources, offering easy dashboard customisation and intuitive resource search
- **Azure PowerShell**: A powerful command-line tool allowing for automation and scripting; example: az vm restart -g MyResourceGroup -n MyVm
- **Azure CLI**: A cross-platform command-line tool that enables managing Azure resources directly from the terminal, which is useful for automating deployments and managing resources at scale

Comparison of Administrator Tools

Azure Portal: Best for visual management and quick configurations; suitable for beginners and visual monitoring

PowerShell and CLI: Ideal for automating tasks, handling complex, repetitive operations and working with bulk resources; CLI and PowerShell scripts can be reused and shared, making them effective for large-scale deployments

Azure Resource Manager (ARM) Templates

Purpose: JSON-based templates that allow you to define resources and infrastructure declaratively

Advantages

- **Consistency**: Templates enable the consistent deployment of resources across environments
- **Reusability**: ARM templates can be saved and reused, promoting efficient scaling
- **Error reduction**: Automates deployment, minimising human error

Template Structure

- **Schema**: Defines the structure of resources in JSON format
- **Parameters**: Includes configurable settings such as VM usernames, passwords or resource sizes

- **Outputs**: Information provided upon deployment completion, often used for referencing deployed resources

Azure Bicep Files

Overview: A language to simplify ARM templates, reducing the complexity of JSON by using a more readable syntax

Features:

- **Simplified syntax**: Easier to write and maintain than ARM templates
- **Modularity**: Allows smaller, reusable modules that can reference each other
- **Automatic dependency detection**: Automatically identifies dependencies between resources
- **Tooling**: Supported by Visual Studio Code with features such as validation, IntelliSense and debugging

Benefits of ARM Templates

Automation and consistency: Simplifies large-scale deployments by ensuring configurations are identical across environments

Modular and linked resources: Allows complex infrastructure to be organised into smaller, manageable modules

Error prevention: Decreases the likelihood of errors by defining infrastructure in code, which can be validated before deployment

Session 2

Understanding Virtual Networks in Azure

Purpose: A virtual network (VNet) provides isolated, secure network environments within Azure, connecting Azure resources and supporting hybrid cloud scenarios

Key Features

- **Isolation**: VNet provides complete control over IP address blocks, DNS settings, security policies and routing.

- **Hybrid connectivity**: VNets can securely connect with on-premises environments, supporting VPN or ExpressRoute connections.

Planning and Creating Virtual Networks

These are the steps in creating VNets:

- **Address space configuration**: Assigns IP address ranges for the VNet
- **Subnet creation**: Ensures logical segmentation within the VNet, allowing different resources to be isolated as needed
- **Flexible deployment**: New VNets can be created during VM deployment or as standalone resources for future expansion

Amazon Container Services

Purpose: Subnets segment the VNet into smaller, logical sections for optimised resource management and security.

Best Practices

- **Unique address ranges**: Each subnet must have a unique IP address range within the VNet.
- **Performance and security**: Segmenting workloads in different subnets can improve security and network performance.

Configuring Public and Private IP Addressing

Private IPs:

- **Usage**: Used for internal communications within Azure and on-premises networks
- **Types**: Assigned dynamically by default but can also be static for predictable addressing

Public IPs:

- **Types**: Available in IPv4 or IPv6, with dynamic and static options
- **Routing options**: Microsoft or internet routing; supports load balancers, VPNs and application gateways

Network Security Groups (NSGs) and Security Rules

NSGs: Used to control inbound and outbound network traffic to Azure resources within a VNet

Security Rules:

- **Configurable options**: Source (IP addresses and ASGs), destination, service and priority levels
- **Evaluation**: Rules are evaluated at the subnet and network interface levels, ensuring comprehensive security coverage

Application Security Groups (ASGs)

Purpose: ASGs allow the logical grouping of VMs based on the application tiers (e.g., web server and database) within a VNet.
Use case: ASGs are used with NSGs to control the traffic flow to and from application components within the VNet.

Azure DNS and Domain Hosting

Azure DNS: Hosts DNS records for custom domains, providing both public and private DNS capabilities
DNS Zones:

- **Public DNS Zones**: Used for external, internet-facing domains
- **Private DNS Zones**: Provides name resolution within and across VNets without relying on external DNS providers

Features:

- **Alias Records**: Dynamically resolve resource names
- **Delegation**: Azure DNS provides specific name server addresses for each DNS zone, verifying domain ownership